## Supporting Strategies to Address Pulmonary Arteriovenous Malformations in Fontan Patients

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### Introduction

The Fontan operation is the final step in a series of palliative surgeries for patients with single-ventricle disease, creating an alternative pathway for pulmonary circulation. This final surgery creates a new structure called the total cavopulmonary circulation (TCPC)[1], where the main pulmonary artery (MPA) is disconnected from the right ventricle and the vena cava is anastomosed to the MPA. Although this circulation initially improves oxygen levels that were lower at birth, if the TCPC design does not distribute hepatic flow evenly to both sides of the lungs then new secondary conditions can arise called pulmonary arteriovenous malformations (PAVMs)[2].

To address PAVMs, a number of interventions have been designed to address uneven hepatic flow in Fontan patients[3]. These interventions can be organized into two categories according to the measured invasiveness: major reconstruction or minimally invasive procedures. Major reconstruction surgeries such as Fontan revisions, Y-grafts, and the implementation of mechanical devices see major anatomical changes that directly impact hepatic flow. Minimally invasive procedures utilize tools like stents to indirectly treat hepatic flow by addressing areas of stenosis or blocking pathways formed by PAVMs[4]. The influence of these interventions has been analyzed in past studies, but a framework to effectively measure and compare each intervention is not available for clinicians to rely on. The aim of this study is to create a framework that can support clinicians in analyzing different interventional strategies that address uneven hepatic flow in Fontan patients.

### Methods

An MRI dataset containing anatomical images and temporal flow information of Fontan patients was used in this study. Five patients' TCPC were selected for this study and subjected to analysis via 3D slicer, a segmentation software. For each patient, the major blood vessels compromising the TCPC were highlighted and segmented: the superior vena cava (SVC), inferior vena cava (IVC), right pulmonary artery (RPA), and the left pulmonary artery (LPA). Each blood vessels contained approximately 20-30 layers of segmentation. These surface models were then transferred to MATLAB for automatic analysis to detect the blood vessel's average cross-sectional area ($cm^2$), centerline of each blood vessel using VMTK[5], and angle of insertion. By analyzing these parameters different interventions strategies could be simulated in MATLAB.

### Results

Below, Figure 1 and Figure 2 demonstrate both minimally and major invasive approaches. Figure 1 demonstrates the use of an adjustable virtual stent to address layers within the IVC with minimal cross-sectional area. Figure 2 represents a potential Fontan revision by alternating the placement of the IVC in the TCPC.

### Conclusion

This preliminary data demonstrates the potential of simulating interventions strategies to improve imbalanced hepatic blood flow within the TCPC. The potential to translate these strategies allows deeper insights and flexibility for clinicians. Further studies will include the calculation of blood flow pre and postoperative to understand the efficacy of these interventions. By providing a framework to simulate these interventions, the doorways to more insightful procedures can be opened leading to more effective procedures.

## References


1. **Anderson, Page A. W., Lynn A. Sleeper, Lynn Mahony, Steven D. Colan, Andrew M. Atz, Roger E. Breitbart, Welton M. Gersony, et al.** 2008. “Contemporary Outcomes after the Fontan Procedure: A Pediatric Heart Network Multicenter Study.” *Journal of the American College of Cardiology* 52 (2): 85–98. https://doi.org/10.1016/j.jacc.2008.01.074.
2. **Kavarana, Minoo N., Jeffrey P. Lamberti, Frank A. Pigula, John E. Mayer Jr., Pedro J. del Nido, and Richard A. Jonas.** 2002. “Pulmonary Arteriovenous Malformations After the Superior Cavopulmonary Shunt: Mechanisms and Clinical Implications.” *The Annals of Thoracic Surgery* 73 (6): 1937–1941. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4410694/
3. **Ohuchi, Hideo.** 2017. “Where Is the ‘Optimal’ Fontan Hemodynamics?” *Korean Circulation Journal* 47 (6): 842–857. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5711675/pdf/kcj-47-842.pdf
4. **Tokunaga, Shigehiko, Hideaki Kado, Yutaka Imoto, Munetaka Masuda, Yuichi Shiokawa, Kouji Fukae, Naoki Fusazaki, Shiro Ishikawa, and Hisataka Yasui.** 2002. “Total Cavopulmonary Connection with an Extracardiac Conduit: Experience with 100 Patients.” *They Annals of Thoracic Surgery* 73 (1): 76–80. https://doi.org/10.1016/S0003-4975(01)03302-1
5. **Antiga L, Piccinelli M, Botti L, Ene-Iordache B, Remuzzi A, Steinman DA**. "An image-based modeling framework for patient-specific computational hemodynamics." Medical & Biological Engineering & Computing. 2008;46(11):1097–1112. https://doi.org/10.1007/s11517-008-0420-1.


### Supplemental Material

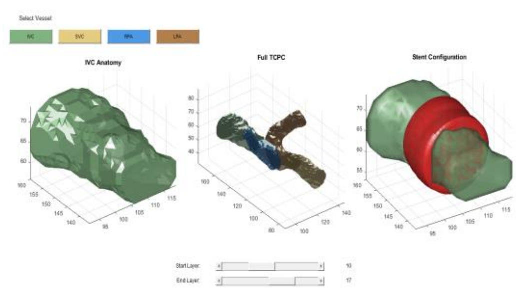


**Figure 1. Virtual Stent Implementation to modify IVC’s cross sectional area**

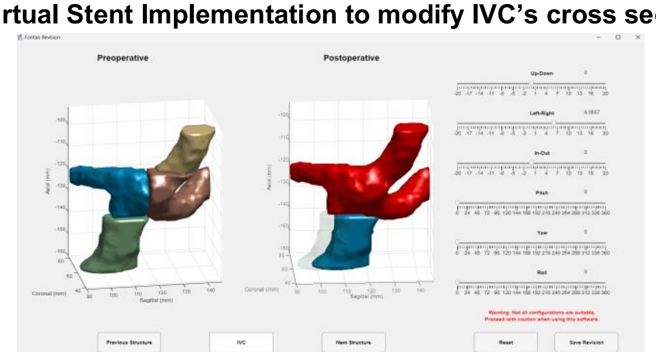


**Figure 2. Fontan Revision Strategy aimed at improved IVC configuration**